

Exercicio

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1 1. Configuração

2 2. Pacotes

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(igraph)
library(ggraph)
library(tidygraph)
library(graphlayouts)
library(pheatmap)
library(corrplot)
library(reshape2)
library(scales)
```

3 3. Introdução

A presente atividade visa apresentar a representação da rede de coocorrência das patologias torácicas.

4 4. Matriz de coocorrência

```
cooc <- matrix(c(
  4212,369,3269,3259,727,585,243,772,1222,221,423,220,495,40,
  369,1094,1060,583,99,108,36,48,169,127,44,51,111,7,
  3269,1060,3959,3990,1244,909,253,995,1287,592,359,188,848,21,
  3259,583,3990,9552,1151,1544,571,943,1220,979,447,345,749,33,
  727,99,1244,1151,2138,894,62,424,602,128,212,115,448,25,
  585,108,909,1544,894,2706,63,340,428,131,115,166,410,10,
  243,36,253,571,62,63,307,34,114,330,21,11,45,2,
  772,48,995,943,424,340,34,2199,222,33,746,80,289,9,
  1222,169,1287,1220,602,428,114,222,1314,162,103,79,251,4,
  221,127,592,979,128,131,330,33,162,634,30,9,64,3,
  423,44,359,447,212,115,21,746,103,30,895,36,151,4,
  220,51,188,345,115,166,11,80,79,9,36,727,176,8,
  495,111,848,749,448,410,45,289,251,64,151,176,1127,8,
  40,7,21,33,25,10,2,9,4,3,4,8,8,110
), nrow = 14, byrow = TRUE)

colnames(cooc) <- rownames(cooc) <- c(
  "Atelectasis",
  "Cardiomegaly",
  "Infiltration",
  "Consolidation",
  "Edema",
  "Emphysema",
  "Fibrosis",
  "Effusion",
  "Pneumonia",
  "Pleural_Thickening",
  "Pneumothorax",
  "Nodule",
  "Mass",
  "Hernia"
)

cooc
```

##	Atelectasis	Cardiomegaly	Infiltration	Consolidation	Edema
## Atelectasis	4212	369	3269	3259	727
## Cardiomegaly	369	1094	1060	583	99
## Infiltration	3269	1060	3959	3990	1244
## Consolidation	3259	583	3990	9552	1151
## Edema	727	99	1244	1151	2138
## Emphysema	585	108	909	1544	894

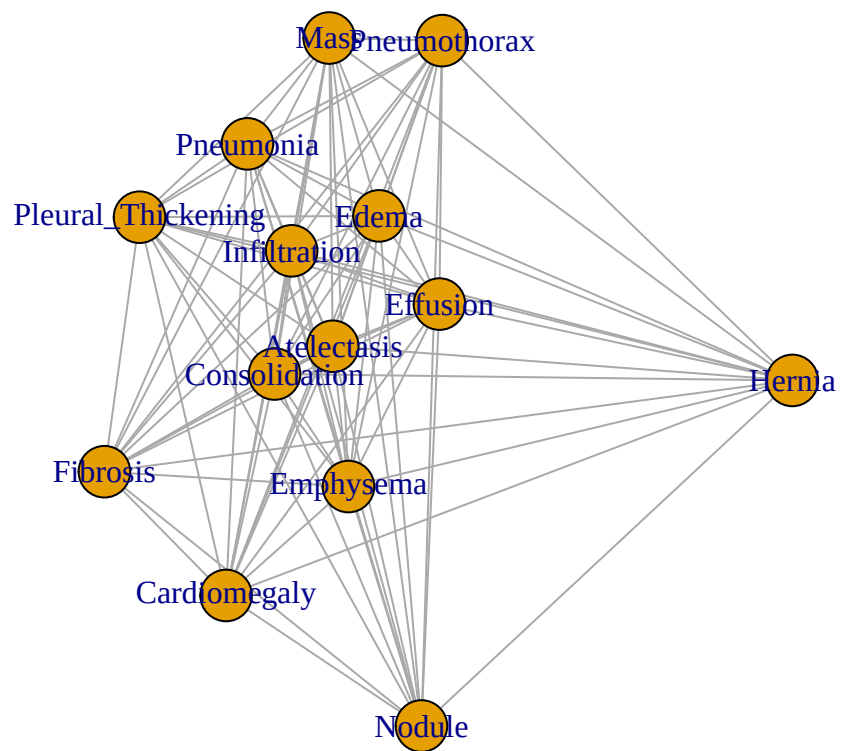
## Fibrosis	243	36	253	571	62
## Effusion	772	48	995	943	424
## Pneumonia	1222	169	1287	1220	602
## Pleural_Thickening	221	127	592	979	128
## Pneumothorax	423	44	359	447	212
## Nodule	220	51	188	345	115
## Mass	495	111	848	749	448
## Hernia	40	7	21	33	25
##	Emphysema	Fibrosis	Effusion	Pneumonia	Pleural_Thickening
## Atelectasis	585	243	772	1222	221
## Cardiomegaly	108	36	48	169	127
## Infiltration	909	253	995	1287	592
## Consolidation	1544	571	943	1220	979
## Edema	894	62	424	602	128
## Emphysema	2706	63	340	428	131
## Fibrosis	63	307	34	114	330
## Effusion	340	34	2199	222	33
## Pneumonia	428	114	222	1314	162
## Pleural_Thickening	131	330	33	162	634
## Pneumothorax	115	21	746	103	30
## Nodule	166	11	80	79	9
## Mass	410	45	289	251	64
## Hernia	10	2	9	4	3
##	Pneumothorax	Nodule	Mass	Hernia	
## Atelectasis	423	220	495	40	
## Cardiomegaly	44	51	111	7	
## Infiltration	359	188	848	21	
## Consolidation	447	345	749	33	
## Edema	212	115	448	25	
## Emphysema	115	166	410	10	
## Fibrosis	21	11	45	2	
## Effusion	746	80	289	9	
## Pneumonia	103	79	251	4	
## Pleural_Thickening	30	9	64	3	
## Pneumothorax	895	36	151	4	
## Nodule	36	727	176	8	
## Mass	151	176	1127	8	
## Hernia	4	8	8	110	

5 5. Construção da rede

A partir da matriz de coocorrência foi construída uma rede ponderada, na qual os nós representam patologias e os pesos das arestas representam a frequência de coocorrência.

```
g_0 <- graph_from_adjacency_matrix(
  cooc,
  mode = "undirected",
  weighted = TRUE,
  diag = FALSE
)

plot(g_0)
```



6 6. Rede com limiar de 75%

Para destacar as associações mais fortes entre as patologias, foi utilizado o percentil 75 dos pesos das arestas.

```
# Pesos das arestas da parte superior da matriz
pesos <- cooc[upper.tri(cooc)]

# Limiar do percentil 75
limiar75 <- quantile(pesos, 0.75)

limiar75
```

```
## 75%
## 584
```

```

# Criar matriz filtrada
cooc75 <- cooc

cooc75[cooc75 < limiar75] <- 0

# Retirar diagonal
diag(cooc75) <- 0

# Criar a rede filtrada
g75 <- graph_from_adjacency_matrix(
  cooc75,
  mode = "undirected",
  weighted = TRUE,
  diag = FALSE
)

# Remover nós isolados
g75 <- delete_vertices(
  g75,
  degree(g75) == 0
)

# Força dos nós
V(g75)$strength <- strength(g75)

```

7 7. Comunidades

```

com75 <- cluster_louvain(g75)

V(g75)$community <- membership(com75)

table(V(g75)$community)

```

```

##
## 1 2
## 9 2

```

8 8. Incidência das patologias por comunidade

```

L75 <- layout_with_fr(
  g75,
  weights = E(g75)$weight
)

cores <- c(
  "#4E79A7",
  "#59A14F",

```

```

"#E15759",
"#F28E2B",
"#B07AA1"
)

plot(
  g75,
  layout = L75,
  vertex.size = rescale(
    V(g75)$strength,
    to = c(15, 35)
  ),
  vertex.color = cores[V(g75)$community],
  vertex.frame.color = "grey20",
  vertex.label.cex = 0.9,
  edge.width = rescale(
    E(g75)$weight,
    to = c(1, 6)
  ),
  edge.color = "grey60"
)

legend(
  "bottomleft",
  legend = paste(
    "Community",
    sort(unique(V(g75)$community))
  ),
  pt.bg = cores[
    sort(unique(V(g75)$community))
  ],
  pch = 21,
  pt.cex = 2,
  bty = "n"
)

```

